

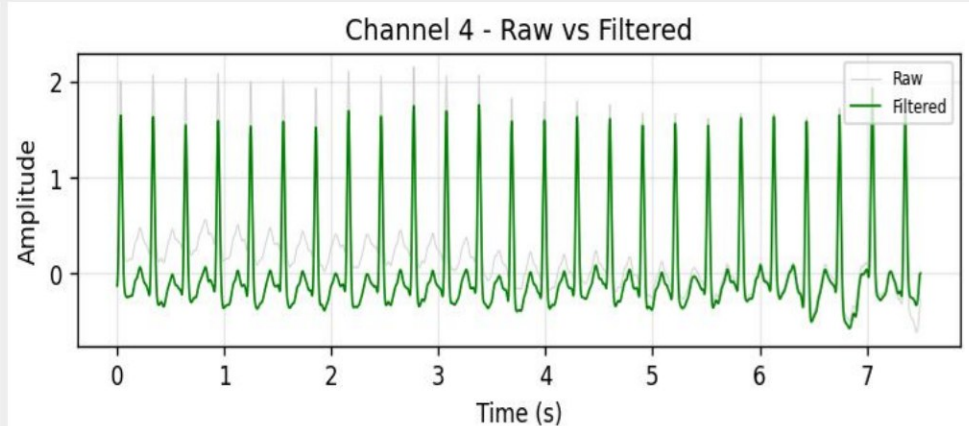
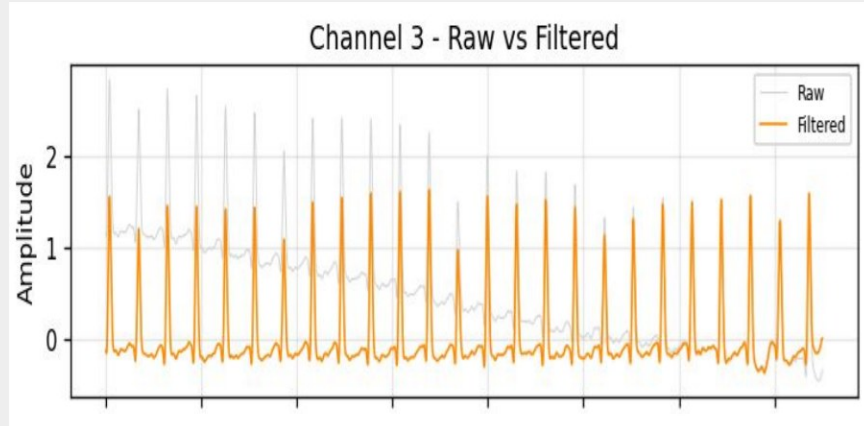
ECG Feature Extraction - AVNRT/AVRT & ZC Prediction

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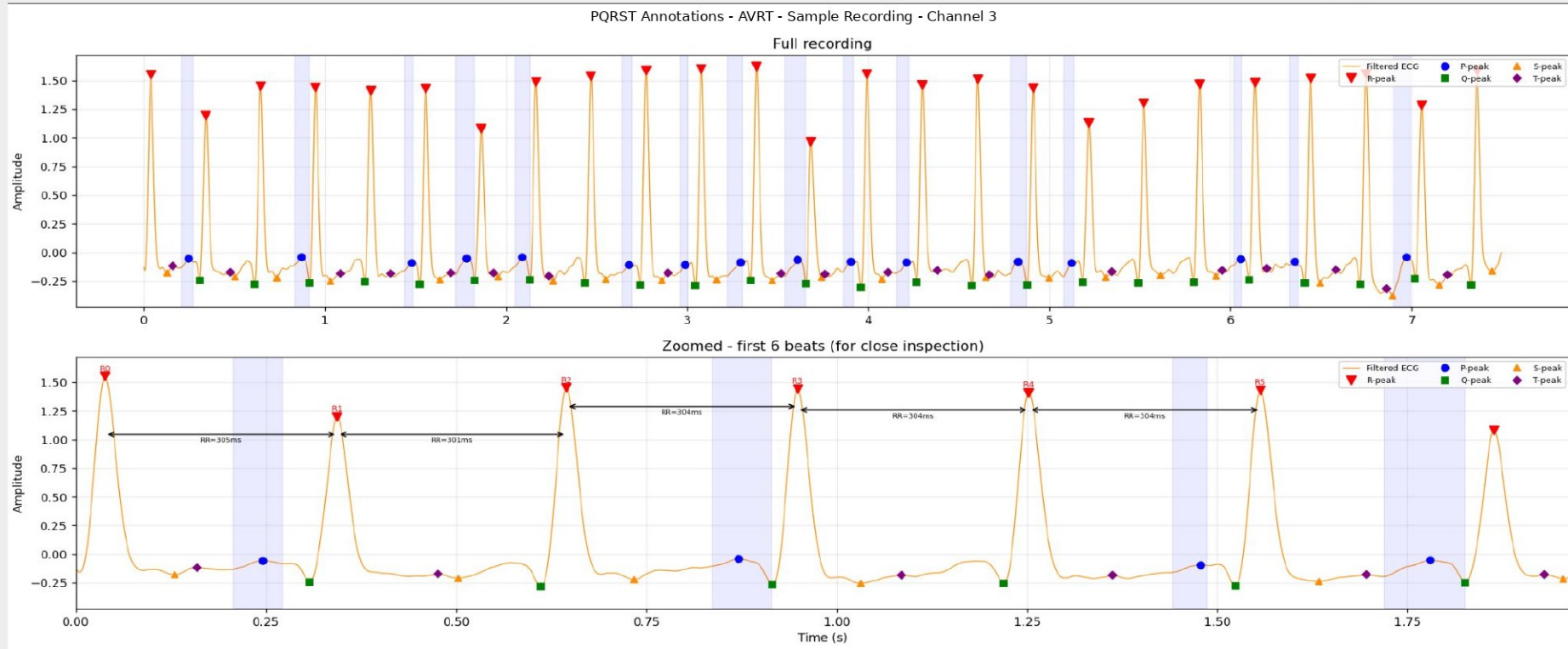
Introduction: ECG Feature Extraction - AVRT/AVNRT

- **Goal:** Extract clinically relevant features from ECG signals and develop ML models for AVRT/AVNRT classification.
- **Dataset And Features Extracted:** 15 patients (8 AVNRT, 7 AVRT) - 46 recordings (10,000 Hz, 7.5 sec)
 - **Heart rate variability** (*pNN50, RMSSD, RR_std, HR_std*) - heartbeat timing variation over time
 - **Poincaré plot** (*SD1, SD2, SD1/SD2 ratio*) - compares each heartbeat gap to next (variability)
 - **QRS shape** (*beat skewness, beat kurtosis, ST slope, P_wave_absent_flag*) - shape of main heartbeat spike (QRS complex)



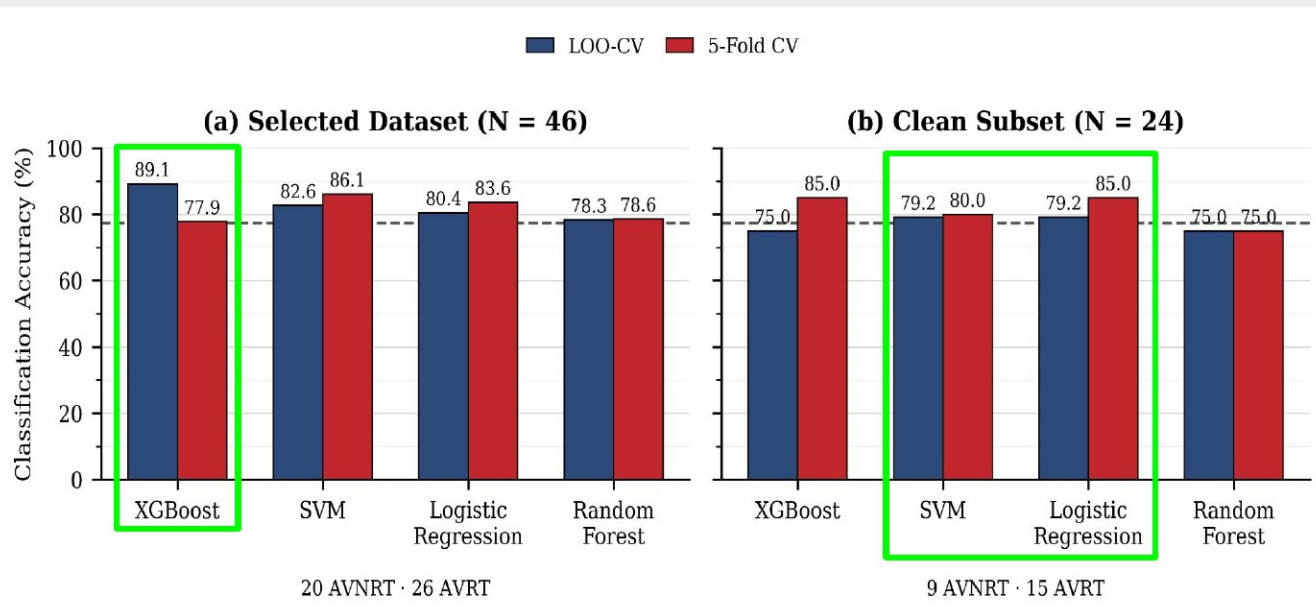
Methodology: Feature Extraction (AVRT & AVNRT) - Channel 3 as sample

- Filtered out noise from raw signal (**Butterworth filter, 0.5–50 Hz**) - **Channels 3 & 4**
- Checked signal quality - signal strength vs. noise ratio (**SNR**) > 0.5
- Detected each heartbeat peak and shape (P, Q, R, S, T points) - **NeuroKit & Pan Tompkins Algorithm**
- Significance test (**Mann-Whitney, $p < 0.05$**) for AVNRT/AVRT classification



Results: AVRT/AVNRT Classification

- **Full Dataset (46 recordings):** **XGBoost** - 89.1% accuracy (LOOCV by leaving one patient out at a time), 5-Fold CV (77.9%)
- **Smaller Dataset (24 recordings):** Confirmed through SNR ratio and visual inspection.
 - Simpler models **(Logistic Regression & SVM)** did best: 75-85% accuracy (LOOCV & 5-Fold CV)
- **Validation:** SHAP > 0.3 across 50% of all features
- **Sanity Check:** Results similar to CNN Paper for 12 lead ECG (77.45 AUC)



Introduction: Adding ECG Features To Zero Crossing (ZC) Prediction *

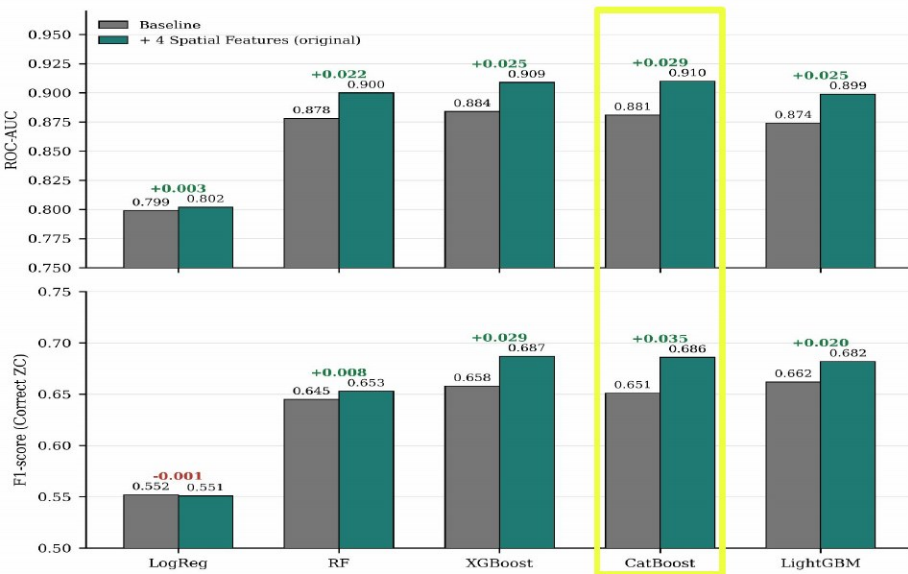
- **Background:** A **Zero Crossing (ZC)** is the point on the strain curve that crosses zero, marking the switch between heart contraction and relaxation (*GT_Label*).
- **Current Zero Crossing (ZC) Dataset:** Contains X & Y coordinates, existing spatial features, *GT_Label* and ZC slope.
- **Goal:** Test whether the **same 11 ECG features used for AVNRT/AVRT** can help **predict whether a ZC point is correct.**
- **Approach:** Add ECG feature values to each ZC point and add spatial features on top of the ones already in the dataset.

*** Paper Reference:** C. Proestaki, C. Ozsoy, M. Tourni, E. S. Silver, L. Liberman, and E. Konofagou, "Intraoperative electromechanical wave imaging for differentiation of AVNRT and AVRT using manual and machine-learning-based atrioventricular activation delays," *Heart Rhythm*, vol. 23, no. 4, p. S546, Apr. 2026, doi: 10.1016/j.hrthm.2026.03.829.

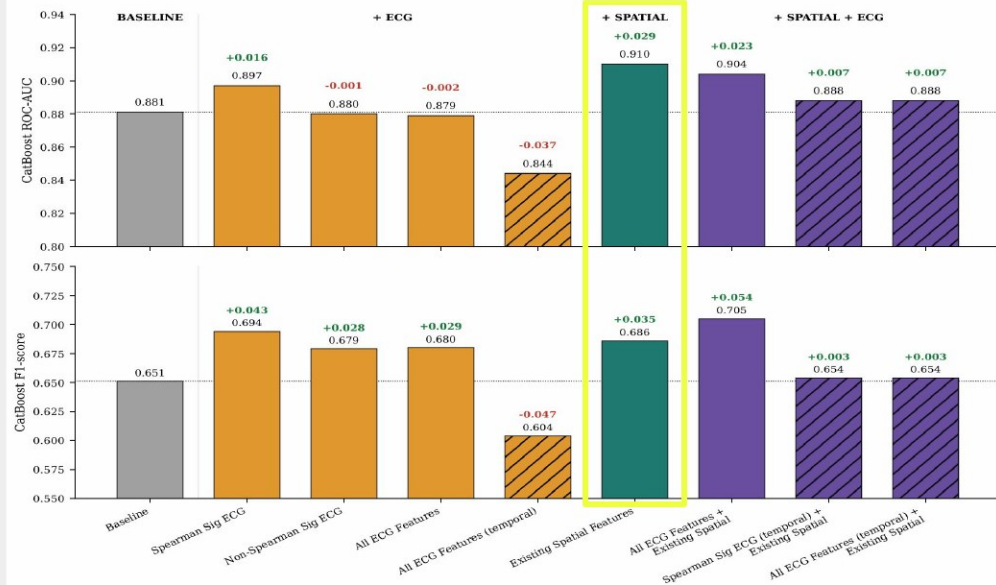
Methodology & Results: Before Spatial Features

- **Testing:** 5 ML models - **CatBoost** performed best overall
- **Validation:** By *patient* (80/20 split: 12 train, 3 test)
- **Metrics:** **ROC-AUC** (how well the model ranks correct vs. incorrect ZCs) and **F1 score** (balance of precision and recall)
- **Observation:** **Adding spatial features** improved CatBoost the most compared to ECG: ROC-AUC rose from 0.881 to 0.910

Cross-Model Comparison: Baseline vs. Spatial Features



ROC-AUC & F1: CatBoost



Results: After Adding More Spatial Features (Spatial Engineering)

- **Enhancement**: Added 7 new spatial features on top of the original 4
- **Best Result**: **CatBoost**: ROC-AUC 0.917, F1 score 0.733
- **Conclusion**: **Spatial features** mattered more than adding extra ECG features

